

Gradient Descent

Exam-Pattern Practice Problems – Fully Worked, Step by Step

ZC416 Mathematical Foundations for ML/DS • Lecture 8 (Topic #9)

All questions use **fresh functions and data** in the style typically asked in quizzes, mid-semester and comprehensive exams. Every problem is numerical: compute, derive, or trace.

#	Skill tested	Difficulty	Marks
1	Critical points + second-derivative test (1-D)	Easy	4
2	Two GD iterations on $f(x, y) = x^2 + 2y^2$	Easy	4
3	Learning-rate decay computations	Easy	4
4	Exact minimizer + GD on a shifted bowl	Easy	4
5	Critical points in 2-D via the Hessian	Moderate	6
6	Quadratic form: fixed step vs. exact line search	Moderate	8
7	Derive the optimal step size γ^* and iterate	Moderate	8
8	Gradient checking by finite differences	Moderate	6
9	Exact line-search step on $x^2 + 4y^2$	Moderate	8
10	GD on a coupled quadratic $x^2 + xy + y^2$	Difficult	8
11	Train linear regression by GD + step-size limit	Difficult	10
12	Binary-search line search: trace + iteration count	Difficult	6

Problem 1 (4 marks) **EASY** **COMPUTATION**

Let $f(x) = 2x^3 - 9x^2 + 12x + 5$.

- (a) Find all stationary points of f . [2]
- (b) Classify each using the second-derivative test and evaluate f there. [1]
- (c) Does f possess a global minimum over $\mathbb{R}$? Justify. [1]

Step-by-step solution

Step 1 (set the gradient to zero). Differentiate and factor:

$$f'(x) = 6x^2 - 18x + 12 = 6(x - 1)(x - 2).$$

So $f'(x) = 0$ at $x = 1$ and $x = 2$.

Step 2 (second-derivative test). Here $f''(x) = 12x - 18$.

$$f''(1) = -6 < 0 \Rightarrow \text{local maximum at } x = 1, \quad f''(2) = +6 > 0 \Rightarrow \text{local minimum at } x = 2.$$

Step 3 (values).

$$f(1) = 2 - 9 + 12 + 5 = 10, \quad f(2) = 16 - 36 + 24 + 5 = 9.$$

Step 4 (global check). As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ (odd degree). The function is unbounded below, so $x = 2$ is only a *local* minimum.

Answer. $x = 1$: local max, $f = 10$. $x = 2$: local min, $f = 9$. No global minimum – a cubic is unbounded.

Intuition & exam tips

The standard 4-marker: factor the derivative (try integer roots first), test curvature, then – the step students skip – check the behaviour at $\pm\infty$ before saying “global”. A cubic never has a global minimum or maximum.

Problem 2 (4 marks) **EASY** COMPUTATION

Apply gradient descent to $f(x, y) = x^2 + 2y^2$ with learning rate $\gamma = 0.1$, starting at $(x_0, y_0) = (4, 2)$.

- (a) Write the update equations for x and y . [1]
- (b) Carry out **two** iterations, reporting the point and f after each. [2]
- (c) Verify that f decreased monotonically. [1]

Step-by-step solution

Step 1 (update equations). The gradient is $\nabla f = (2x, 4y)$, so

$$x_{k+1} = x_k - \gamma (2x_k) = 0.8 x_k, \quad y_{k+1} = y_k - \gamma (4y_k) = 0.6 y_k.$$

Step 2 (iterate – keep a table).

k	point (x_k, y_k)	gradient $(2x_k, 4y_k)$	$f(x_k, y_k)$
0	(4, 2)	(8, 8)	$16 + 8 = 24$
1	$(4, 2) - 0.1 (8, 8) = (3.2, 1.2)$	(6.4, 4.8)	$10.24 + 2.88 = 13.12$
2	$(3.2, 1.2) - 0.1 (6.4, 4.8) = (2.56, 0.72)$	–	$6.5536 + 1.0368 = 7.5904$

Step 3 (monotone decrease).

$$24 \geq 13.12 \geq 7.5904. \checkmark$$

Both coordinates head toward the minimizer $(0, 0)$; y converges faster since its factor 0.6 is smaller than 0.8.

Answer. $(x_1, y_1) = (3.2, 1.2)$ with $f = 13.12$; $(x_2, y_2) = (2.56, 0.72)$ with $f = 7.5904$; f strictly decreasing.

Intuition & exam tips

“Perform k iterations of GD” is the most frequently asked computation on this topic. Always work in a table – point, gradient, new point, f – so no bookkeeping marks are lost. Each coordinate here shrinks by its own constant factor $(1 - \gamma \times \text{curvature})$; the same observation powers Problem 11(c).

Problem 3 (4 marks) **EASY** COMPUTATION

Training starts with $\alpha_0 = 0.8$ and decay constant $k = 0.05$.

- (a) Compute α_t at $t = 10$ and $t = 20$ under exponential decay $\alpha_t = \alpha_0 e^{-kt}$ and under inverse decay $\alpha_t = \alpha_0 / (1 + kt)$. [2]
- (b) Under step decay that divides α by 3 every 5 epochs, find α during epoch $t = 12$. [1]
- (c) At what epoch does *inverse* decay first reach $\alpha_0/2$? [1]

Step-by-step solution**Step 1 (exponential vs inverse).**

t	$\alpha_t = 0.8 e^{-0.05t}$	$\alpha_t = 0.8/(1 + 0.05t)$
10	$0.8 e^{-0.5} = 0.4852$	$0.8/1.5 = 0.5333$
20	$0.8 e^{-1} = 0.2943$	$0.8/2 = 0.4000$

Exponential decays faster (geometric vs harmonic shrinkage).

Step 2 (step decay). Divisions happen at $t = 5$ and $t = 10$, so epoch 12 sits in the third block:

$$\alpha = \frac{0.8}{3^2} = \frac{0.8}{9} \approx 0.0889.$$

Step 3 (half-life of inverse decay).

$$\frac{\alpha_0}{1 + kt} = \frac{\alpha_0}{2} \Rightarrow 1 + 0.05t = 2 \Rightarrow t = \frac{1}{k} = 20 \text{ epochs.}$$

Answer. (a) 0.4852, 0.2943 (exponential) vs 0.5333, 0.4 (inverse); (b) 0.0889; (c) $t = 1/k = 20$.

Intuition & exam tips

Three plug-in formulas – but part (c)'s pattern ($t = 1/k$ halves inverse decay; the same t multiplies exponential decay by $1/e$) is what separates a full-marks answer. In the update $w_{n+1} = w_n - \alpha_t \nabla J$, the counter t is measured in epochs, so α is constant within an epoch.

Problem 4 (4 marks) EASY COMPUTATION

Let $f(x, y) = x^2 + y^2 - 2x - 4y + 5$.

- (a) Find the exact minimizer (x^*, y^*) and the minimum value. [1]
 (b) Run two GD iterations from $(0, 0)$ with $\gamma = 0.25$, reporting f each time. [3]

Step-by-step solution**Step 1 (complete the square).**

$$f(x, y) = (x - 1)^2 + (y - 2)^2,$$

so the minimizer is $(x^*, y^*) = (1, 2)$ with $f^* = 0$. (Equivalently: $\nabla f = (2x - 2, 2y - 4) = (0, 0)$ gives the same point.)

Step 2 (iteration 1). Gradient at $(0, 0)$: $\nabla f = (-2, -4)$.

$$(x_1, y_1) = (0, 0) - 0.25(-2, -4) = (0.5, 1), \quad f(x_1, y_1) = (-0.5)^2 + (-1)^2 = 1.25.$$

(Started from $f(0, 0) = 5$.)

Step 3 (iteration 2). Gradient at $(0.5, 1)$: $\nabla f = (-1, -2)$.

$$(x_2, y_2) = (0.5, 1) - 0.25(-1, -2) = (0.75, 1.5), \quad f(x_2, y_2) = (-0.25)^2 + (-0.5)^2 = 0.3125.$$

So $f : 5 \rightarrow 1.25 \rightarrow 0.3125$, closing in on $(1, 2)$.

Answer. (a) $(x^*, y^*) = (1, 2)$, $f^* = 0$; (b) $(0.5, 1)$ then $(0.75, 1.5)$, with $f : 5 \rightarrow 1.25 \rightarrow 0.3125$.

Intuition & exam tips

Completing the square first tells you the destination, so every GD iterate can be sanity-checked against it – here each iterate lands exactly halfway closer to $(1, 2)$. If an exam asks “how many more iterations until $f \leq 0.01$?”, continue the visible pattern: f shrinks by factor 4 per step.

Problem 5 (6 marks) MODERATE COMPUTATION

Let $f(x, y) = x^3 - 3x + y^2 - 2y$.

- (a) Find all critical points (where $\nabla f = \mathbf{0}$). [2]
- (b) Classify each using the Hessian matrix. [3]
- (c) Evaluate f at each critical point and state which one gradient descent can converge to. [1]

Step-by-step solution

Step 1 (solve $\nabla f = \mathbf{0}$).

$$\nabla f = (3x^2 - 3, 2y - 2) = (0, 0) \Rightarrow x = \pm 1, \quad y = 1.$$

Critical points: $P_1 = (1, 1)$ and $P_2 = (-1, 1)$.

Step 2 (Hessian test).

$$H(x, y) = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}.$$

At $P_1 = (1, 1)$: $H = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$ – both eigenvalues $(6, 2)$ positive $\Rightarrow$ positive definite $\Rightarrow$ **local minimum**.

At $P_2 = (-1, 1)$: $H = \begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}$ – eigenvalues $(-6, 2)$ of mixed sign $\Rightarrow$ indefinite $\Rightarrow$ **saddle point** (neither min nor max).

Step 3 (values and the reachable point).

$$f(1, 1) = 1 - 3 + 1 - 2 = -3, \quad f(-1, 1) = -1 + 3 + 1 - 2 = 1.$$

GD (with reasonable starts) converges to the local minimum $P_1 = (1, 1)$; at the saddle P_2 the gradient is zero but any perturbation in the x -direction escapes it. Note f has no global minimum: $f \rightarrow -\infty$ as $x \rightarrow -\infty$.

Answer. Critical points $(1, 1)$ and $(-1, 1)$; $(1, 1)$ is a local min ($f = -3$), $(-1, 1)$ is a saddle ($f = 1$); GD targets $(1, 1)$.

Intuition & exam tips

The 2-D version of Problem 1, and just as common in papers. For a *diagonal* Hessian the eigenvalues are simply the diagonal entries: all positive = min, all negative = max, mixed signs = saddle. Saddles matter for GD because the gradient vanishes there too – “ $\nabla f = 0$ ” alone never proves you found a minimum.

Problem 6 (8 marks) MODERATE COMPUTATION + DERIVATION

Let $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top A\mathbf{x} - \mathbf{b}^\top \mathbf{x}$ with

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

- (a) Write ∇f and find the exact minimizer $\mathbf{x}^*$ and $f(\mathbf{x}^*)$. [3]
- (b) From $\mathbf{x}_0 = (0, 0)^\top$ take one step with fixed $\gamma = 0.1$ and evaluate f . [2]
- (c) Show that the **exact line-search** step length from $\mathbf{x}$ along $-\mathbf{g}$ (where $\mathbf{g} = \nabla f(\mathbf{x})$) is $\gamma^* = \frac{\mathbf{g}^\top \mathbf{g}}{\mathbf{g}^\top A \mathbf{g}}$, and use it from the same $\mathbf{x}_0$. Compare with (b). [3]

Step-by-step solution

Step 1 (gradient and minimizer). For this quadratic,

$$\nabla f(\mathbf{x}) = A\mathbf{x} - \mathbf{b}.$$

Setting $A\mathbf{x} = \mathbf{b}$ gives the linear system

$$4x + y = 1, \quad x + 3y = 2.$$

From the first equation, $y = 1 - 4x$. Substitute into the second:

$$x + 3(1 - 4x) = 2 \Rightarrow -11x = -1 \Rightarrow x = \frac{1}{11}, \quad y = 1 - \frac{4}{11} = \frac{7}{11}.$$

So

$$\mathbf{x}^* = \left(\frac{1}{11}, \frac{7}{11} \right) \approx (0.091, 0.636).$$

(A is positive definite – $\det A = 11 > 0$ and $4 > 0$ – so this is the unique global minimum.)

Step 2 (minimum value). Because $A\mathbf{x}^* = \mathbf{b}$, we have $\mathbf{x}^{*\top} A\mathbf{x}^* = \mathbf{b}^\top \mathbf{x}^*$, so

$$f(\mathbf{x}^*) = \frac{1}{2}\mathbf{b}^\top \mathbf{x}^* - \mathbf{b}^\top \mathbf{x}^* = -\frac{1}{2}\mathbf{b}^\top \mathbf{x}^* = -\frac{1}{2}\left(\frac{1}{11} + \frac{14}{11}\right) = -\frac{15}{22} \approx -0.682.$$

Step 3 (one fixed step, $\gamma = 0.1$). At the origin:

$$\mathbf{g}_0 = \nabla f(\mathbf{0}) = -\mathbf{b} = (-1, -2)^\top.$$

$$\mathbf{x}_1 = \mathbf{0} - 0.1(-1, -2)^\top = (0.1, 0.2)^\top.$$

Evaluate $f(\mathbf{x}_1)$ in pieces:

$$A\mathbf{x}_1 = (0.6, 0.7)^\top, \quad \mathbf{x}_1^\top A\mathbf{x}_1 = 0.06 + 0.14 = 0.2, \quad \mathbf{b}^\top \mathbf{x}_1 = 0.1 + 0.4 = 0.5.$$

$$f(\mathbf{x}_1) = \frac{1}{2}(0.2) - 0.5 = -0.4.$$

Step 4 (derive γ^*). Let $h(\gamma) = f(\mathbf{x} - \gamma\mathbf{g})$. By the chain rule and $\nabla f(\mathbf{z}) = A\mathbf{z} - \mathbf{b}$:

$$h'(\gamma) = -\mathbf{g}^\top \nabla f(\mathbf{x} - \gamma\mathbf{g}) = -\mathbf{g}^\top (\underbrace{A\mathbf{x} - \mathbf{b}}_{=\mathbf{g}} - \gamma A\mathbf{g}) = -\mathbf{g}^\top \mathbf{g} + \gamma \mathbf{g}^\top A\mathbf{g}.$$

Setting $h'(\gamma) = 0$:

$$\gamma^* = \frac{\mathbf{g}^\top \mathbf{g}}{\mathbf{g}^\top A \mathbf{g}}$$

(a true minimum of h , since $h'' = \mathbf{g}^\top A \mathbf{g} > 0$ for positive-definite A).

Step 5 (apply γ^* at $\mathbf{x}_0$). With $\mathbf{g}_0 = (-1, -2)^\top$:

$$\mathbf{g}_0^\top \mathbf{g}_0 = 1 + 4 = 5, \quad A \mathbf{g}_0 = (-6, -7)^\top, \quad \mathbf{g}_0^\top A \mathbf{g}_0 = 6 + 14 = 20.$$

$$\gamma^* = \frac{5}{20} = 0.25 \quad \Rightarrow \quad \mathbf{x}_1^{LS} = \mathbf{0} - 0.25(-1, -2)^\top = (0.25, 0.5)^\top.$$

Evaluate:

$$A \mathbf{x}_1^{LS} = (1.5, 1.75)^\top, \quad \mathbf{x}_1^\top A \mathbf{x}_1 = 0.375 + 0.875 = 1.25, \quad \mathbf{b}^\top \mathbf{x}_1 = 1.25,$$

$$f(\mathbf{x}_1^{LS}) = \frac{1}{2}(1.25) - 1.25 = -0.625.$$

Step 6 (compare).

	f after one step
fixed $\gamma = 0.1$	-0.400
exact line search $\gamma^* = 0.25$	-0.625
optimum f^*	-0.682

Same direction, better length: the line-search step covers most of the way to the optimum in one move.

Answer. (a) $\mathbf{x}^* = (1/11, 7/11)$, $f^* = -15/22 \approx -0.682$; (b) $\mathbf{x}_1 = (0.1, 0.2)$, $f = -0.4$; (c) $\gamma^* = \mathbf{g}^\top \mathbf{g} / \mathbf{g}^\top A \mathbf{g} = 0.25$, $\mathbf{x}_1^{LS} = (0.25, 0.5)$, $f = -0.625$.

Intuition & exam tips

$\gamma^* = \mathbf{g}^\top \mathbf{g} / \mathbf{g}^\top A \mathbf{g}$ is the line-search formula examiners expect for quadratics – the 3-line derivation in Step 4 is the full-credit version; reproduce it rather than quoting. Watch the sign convention: with $-\mathbf{b}^\top \mathbf{x}$ in f , the gradient is $A \mathbf{x} - \mathbf{b}$ and the optimum solves $A \mathbf{x} = \mathbf{b}$; papers switch freely between $\pm \mathbf{b}$ versions.

Problem 7 (8 marks) MODERATE DERIVATION + COMPUTATION

Gradient descent is applied to $f(x, y) = 3x^2 + y^2$ starting at $(x_0, y_0) = (1, 3)$, choosing at each step the γ that minimizes $h(\gamma) = f((x, y) - \gamma \nabla f(x, y))$.

(a) Show that $h(\gamma) = 3(1 - 6\gamma)^2 x^2 + (1 - 2\gamma)^2 y^2$ and derive $\gamma^* = \frac{9x^2 + y^2}{54x^2 + 2y^2}$. [5]

(b) Evaluate γ^* at $(1, 3)$; compute the new point and the new value of f . [3]

Step-by-step solution

Step 1 (build the one-variable function h). The gradient is $\nabla f = (6x, 2y)$, so the trial point is

$$(x - \gamma 6x, y - \gamma 2y) = ((1 - 6\gamma)x, (1 - 2\gamma)y).$$

Substituting into f :

$$h(\gamma) = 3(1 - 6\gamma)^2x^2 + (1 - 2\gamma)^2y^2.$$

Step 2 (minimize h). Differentiate with respect to γ and set to zero:

$$h'(\gamma) = -36x^2(1 - 6\gamma) - 4y^2(1 - 2\gamma) = 0.$$

Expand and collect the γ terms:

$$36x^2 + 4y^2 = \gamma(216x^2 + 8y^2) \Rightarrow \gamma^* = \frac{36x^2 + 4y^2}{216x^2 + 8y^2} = \frac{9x^2 + y^2}{54x^2 + 2y^2}.$$

(It is a minimum because $h'' = 216x^2 + 8y^2 > 0$.)

Step 3 (evaluate at $(1, 3)$).

$$\gamma^* = \frac{9(1) + 9}{54(1) + 2(9)} = \frac{18}{72} = \frac{1}{4}.$$

Step 4 (new point and new value).

$$x_1 = (1 - 6 \cdot \frac{1}{4})(1) = -\frac{1}{2}, \quad y_1 = (1 - 2 \cdot \frac{1}{4})(3) = \frac{3}{2}.$$

$$f(x_0, y_0) = 3 + 9 = 12 \quad \longrightarrow \quad f\left(-\frac{1}{2}, \frac{3}{2}\right) = 3 \cdot \frac{1}{4} + \frac{9}{4} = 3.$$

One optimally sized step removed 75% of the objective value.

Answer. (a) $\gamma^* = \frac{9x^2 + y^2}{54x^2 + 2y^2}$; (b) $\gamma^* = \frac{1}{4}$ at $(1, 3)$, new point $(-\frac{1}{2}, \frac{3}{2})$, $f : 12 \rightarrow 3$.

Intuition & exam tips

The safe routine for any $f = ax^2 + by^2$: (1) write the trial point with factors $(1 - c_i\gamma)$, where c_i is the gradient coefficient of that coordinate; (2) differentiate h ; (3) solve the *linear* equation for γ . Re-derive every time – the derivation itself carries most of the marks, and memorized formulas break when the exam swaps coefficients.

Problem 8 (6 marks) MODERATE COMPUTATION

Your code computes gradients of $J(w_1, w_2) = w_1^3 + w_1w_2^2$, and you audit it at $(2, 1)$ with $\Delta = 0.1$.

- (a) Compute the analytic gradient at $(2, 1)$. [2]
- (b) Estimate both partials with the forward difference $(J(\dots, w_i + \Delta, \dots) - J(\dots))/\Delta$; report the absolute errors. [2]
- (c) Re-estimate $\partial J/\partial w_1$ with the central difference and compare error orders. [2]

Step-by-step solution

Step 1 (analytic gradient).

$$\frac{\partial J}{\partial w_1} = 3w_1^2 + w_2^2 \Big|_{(2,1)} = 12 + 1 = 13, \quad \frac{\partial J}{\partial w_2} = 2w_1w_2 \Big|_{(2,1)} = 4.$$

Step 2 (forward differences). Base value: $J(2, 1) = 8 + 2 = 10$.

For w_1 : $J(2.1, 1) = 9.261 + 2.1 = 11.361$, so

$$\frac{11.361 - 10}{0.1} = 13.61 \quad (\text{error} = 0.61).$$

For w_2 : $J(2, 1.1) = 8 + 2(1.21) = 10.42$, so

$$\frac{10.42 - 10}{0.1} = 4.20 \quad (\text{error} = 0.20).$$

Step 3 (central difference for w_1). $J(1.9, 1) = 6.859 + 1.9 = 8.759$:

$$\frac{J(2.1, 1) - J(1.9, 1)}{2(0.1)} = \frac{11.361 - 8.759}{0.2} = 13.01 \quad (\text{error} = 0.01).$$

Forward error is $O(\Delta)$ (about 0.6 here); central error is $O(\Delta^2)$ (about $0.01 \approx \Delta^2$) – the symmetric evaluations cancel the leading Taylor error term. This audit matters because a wrong gradient silently corrupts *every* GD step; checking a few sampled coordinates is the standard cheap safeguard.

Answer. Analytic (13, 4); forward (13.61, 4.20), errors (0.61, 0.20); central 13.01, error 0.01.

Intuition & exam tips

With a deliberately large $\Delta = 0.1$ the error orders become visible in your own arithmetic: shrink Δ tenfold and the forward error drops tenfold, the central error a hundredfold. If asked “why not $\Delta = 10^{-15}$?” – floating-point round-off then dominates the quotient.

Problem 9 (8 marks) MODERATE DERIVATION + COMPUTATION

Steepest descent with **exact line search** is applied to $J(x, y) = x^2 + 4y^2$ at $\mathbf{w}_0 = (2, 1)$.

- Write the descent direction at $\mathbf{w}_0$ and derive the optimal step γ^* by minimizing $h(\gamma) = J(\mathbf{w}_0 - \gamma \nabla J(\mathbf{w}_0))$. [5]
- Compute $\mathbf{w}_1$ and the drop in J . [3]

Step-by-step solution

Step 1 (direction at $\mathbf{w}_0$). $\nabla J = (2x, 8y)$, so

$$\nabla J(\mathbf{w}_0) = (4, 8), \quad \text{descent direction} = -(4, 8).$$

Step 2 (build h). The ray is $(2 - 4\gamma, 1 - 8\gamma)$, hence

$$h(\gamma) = (2 - 4\gamma)^2 + 4(1 - 8\gamma)^2.$$

Step 3 (minimize h).

$$h'(\gamma) = 2(2 - 4\gamma)(-4) + 8(1 - 8\gamma)(-8) = -8(2 - 4\gamma) - 64(1 - 8\gamma).$$

Expand:

$$h'(\gamma) = -16 + 32\gamma - 64 + 512\gamma = 544\gamma - 80.$$

Set to zero:

$$\gamma^* = \frac{80}{544} = \frac{5}{34} \approx 0.147 \quad (h'' = 544 > 0, \text{ a minimum}). \checkmark$$

Step 4 (new point and drop in J).

$$\mathbf{w}_1 = \left(2 - 4 \cdot \frac{5}{34}, 1 - 8 \cdot \frac{5}{34} \right) = \left(\frac{48}{34}, -\frac{6}{34} \right) = \left(\frac{24}{17}, -\frac{3}{17} \right) \approx (1.412, -0.176).$$

Values:

$$J(\mathbf{w}_0) = 4 + 4 = 8, \quad J(\mathbf{w}_1) = \left(\frac{24}{17} \right)^2 + 4 \left(\frac{3}{17} \right)^2 = \frac{576 + 36}{289} = \frac{36}{17} \approx 2.118.$$

The single optimal step removed about 74% of the objective.

Answer. $\gamma^* = 5/34$; $\mathbf{w}_1 = (24/17, -3/17) \approx (1.41, -0.18)$; $J : 8 \rightarrow 36/17 \approx 2.12$.

Intuition & exam tips

Same recipe as Problem 7, dressed in a different quadratic – which is exactly how papers vary it. Note the y -coordinate overshoot past 0 (to $-3/17$): on an elongated bowl even the *optimal* step along the gradient overshoots in the stiff direction, which is why steepest descent zig-zags on such functions.

Problem 10 (8 marks) **DIFFICULT** **COMPUTATION**

Consider the **coupled** quadratic $f(x, y) = x^2 + xy + y^2$.

- Find the minimizer analytically and confirm it is a minimum via the Hessian. [3]
- From $(x_0, y_0) = (2, 0)$ with $\gamma = 0.2$, perform **two** GD iterations, reporting the gradient, the new point and f each time. [4]
- Explain (with one computed number) why the coordinates can no longer be treated independently. [1]

Step-by-step solution

Step 1 (analytic minimizer).

$$\nabla f = (2x + y, x + 2y) = (0, 0).$$

From the two equations: $y = -2x$ and $x = -2y$; substituting gives $x = 4x \Rightarrow x = 0$, then $y = 0$. So the only critical point is $(0, 0)$, with $f(0, 0) = 0$. Hessian:

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \text{eigenvalues } 2 \pm 1 = \{3, 1\} > 0 \Rightarrow \text{positive definite} \Rightarrow \text{global minimum at } (0, 0).$$

Step 2 (iteration 1). At $(2, 0)$:

$$\nabla f(2, 0) = (2(2) + 0, 2 + 0) = (4, 2), \quad f(2, 0) = 4.$$

$$(x_1, y_1) = (2, 0) - 0.2(4, 2) = (1.2, -0.4).$$

$$f(1.2, -0.4) = 1.44 + (1.2)(-0.4) + 0.16 = 1.44 - 0.48 + 0.16 = 1.12.$$

Step 3 (iteration 2). At $(1.2, -0.4)$:

$$\nabla f = (2(1.2) - 0.4, 1.2 + 2(-0.4)) = (2.0, 0.4).$$

$$(x_2, y_2) = (1.2, -0.4) - 0.2(2.0, 0.4) = (0.8, -0.48).$$

$$f(0.8, -0.48) = 0.64 - 0.384 + 0.2304 = 0.4864.$$

So $f : 4 \rightarrow 1.12 \rightarrow 0.4864$, heading toward $(0, 0)$.

Step 4 (why no decoupling). The cross-term xy makes each partial depend on *both* variables: e.g. at $(2, 0)$ the y -gradient is $x + 2y = 2 \neq 0$ even though y is already at its “own” optimal value 0 – the step drags y to -0.4 because of x . Updates of the form $x_{k+1} = (\text{const})x_k$ no longer exist; the coupling is exactly the off-diagonal 1 in H .

Answer. (a) minimizer $(0, 0)$, H has eigenvalues $\{3, 1\}$ – positive definite; (b) $(1.2, -0.4)$ with $f = 1.12$, then $(0.8, -0.48)$ with $f = 0.4864$; (c) cross-term: at $(2, 0)$, $\partial f / \partial y = 2 \neq 0$ despite $y = 0$.

Intuition & exam tips

Most practice happens on separable bowls ($ax^2 + by^2$); exams often add the xy term precisely to test whether you differentiate carefully ($\partial f / \partial x = 2x + y$, not $2x$). Everything else – the update rule, the table discipline – is unchanged. The eigenvalue fact used here (eigenvalues of $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ are $a \pm b$) is worth memorizing.

Problem 11 (10 marks) **DIFFICULT** **COMPUTATION**

The line $\hat{y} = ax + b$ is fitted to

$$\begin{array}{c|cccc} x & 0 & 1 & 2 & 3 \\ y & 1 & 3 & 5 & 7 \end{array} \quad \text{by minimizing} \quad L(a, b) = \sum_{i=1}^4 (y_i - (ax_i + b))^2.$$

- (a) Find the exact minimizer (a^*, b^*) via the normal equations and interpret L^* . [3]
- (b) From $(a_0, b_0) = (0, 0)$ with $\gamma = 0.05$, perform **two** GD iterations; report the gradient, the parameters and L after each. [5]
- (c) The Hessian of L is $H = 2 \begin{pmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{pmatrix}$, with eigenvalues $\lambda \approx 33.6$ and ≈ 2.4 . Compute the largest usable learning rate $\gamma_{\max} = 2/\lambda_{\max}$ and use it to explain the path seen in (b). [2]

Step-by-step solution

Step 1 (the needed sums).

$$\sum x_i = 6, \quad \sum y_i = 16, \quad \sum x_i^2 = 14, \quad \sum x_i y_i = 0 + 3 + 10 + 21 = 34, \quad n = 4.$$

Step 2 (normal equations).

$$a^* = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} = \frac{4(34) - 6(16)}{4(14) - 36} = \frac{136 - 96}{20} = 2, \quad b^* = \frac{\sum y_i - a^* \sum x_i}{n} = \frac{16 - 12}{4} = 1.$$

Check the fit: $y = 2x + 1$ reproduces all four points exactly, so $L^* = 0$ – the data are perfectly linear.

Step 3 (GD iteration 1).

The partial derivatives (with residuals $r_i = y_i - (ax_i + b)$):

$$\frac{\partial L}{\partial a} = -2 \sum x_i r_i, \quad \frac{\partial L}{\partial b} = -2 \sum r_i.$$

At $(0, 0)$: residuals $r = (1, 3, 5, 7)$.

$$\sum x_i r_i = 0+3+10+21 = 34, \quad \sum r_i = 16 \Rightarrow \nabla L = (-68, -32), \quad L_0 = 1+9+25+49 = 84.$$

Update:

$$(a_1, b_1) = (0, 0) - 0.05(-68, -32) = (3.4, 1.6).$$

New predictions and residuals:

$$\hat{y} = (1.6, 5.0, 8.4, 11.8), \quad r = (-0.6, -2.0, -3.4, -4.8),$$

$$L_1 = 0.36 + 4 + 11.56 + 23.04 = 38.96.$$

Step 4 (GD iteration 2). From the residuals above:

$$\sum x_i r_i = 0 - 2.0 - 6.8 - 14.4 = -23.2, \quad \sum r_i = -10.8 \Rightarrow \nabla L = (46.4, 21.6).$$

Update:

$$(a_2, b_2) = (3.4, 1.6) - 0.05(46.4, 21.6) = (1.08, 0.52).$$

New predictions and residuals:

$$\hat{y} = (0.52, 1.60, 2.68, 3.76), \quad r = (0.48, 1.40, 2.32, 3.24),$$

$$L_2 = 0.2304 + 1.96 + 5.3824 + 10.4976 \approx 18.07.$$

Pattern: a jumped *past* the target 2 (to 3.4), then back below it (to 1.08), yet $L : 84 \rightarrow 38.96 \rightarrow 18.07$ keeps falling.

Step 5 (stability limit).

$$\gamma_{\max} = \frac{2}{\lambda_{\max}} \approx \frac{2}{33.6} \approx 0.0595.$$

Our $\gamma = 0.05$ is just below this threshold. The dominant error mode is multiplied by $1 - \gamma\lambda_{\max} \approx 1 - 0.05(33.6) = -0.68$ each iteration: magnitude < 1 , so the loss decreases, but *negative*, so the parameters flip from one side of the optimum to the other – exactly the overshoot-and-return seen in (b). A smaller γ (say 0.02) would descend smoothly but more slowly.

Answer. (a) $(a^*, b^*) = (2, 1)$ with $L^* = 0$; (b) $(3.4, 1.6)$, $L = 38.96$, then $(1.08, 0.52)$, $L \approx 18.07$ – oscillating but improving; (c) $\gamma_{\max} \approx 0.0595$; since $1 - \gamma\lambda_{\max} \approx -0.68$ is negative, iterates alternate sides of the optimum.

Intuition & exam tips

The complete exam story in one question: normal equations give the destination, GD gives the journey, the Hessian explains the driving style. The reusable diagnostic is the number $1 - \gamma\lambda$: its *magnitude* sets the speed, its *sign* tells you whether the approach is smooth or oscillating.

Problem 12 (6 marks) DIFFICULT ALGORITHM TRACE

A line search must minimize the unimodal $h(\alpha) = \alpha^2 - 5\alpha + 8$ over $[0, \alpha_{\max}] = [0, 4]$.

- (a) Trace **three** iterations of binary search: at each midpoint m decide from the sign of the slope (compare $h(m)$ with $h(m + \epsilon)$, $\epsilon = 10^{-6}$) whether the minimum lies left or right, and give the new interval. [4]
- (b) What is the bracket width after k iterations, and how many iterations guarantee width ≤ 0.01 ? [2]

Step-by-step solution

Step 1 (iteration 1). Interval $[0, 4]$, midpoint $m = 2$:

$$h'(\alpha) = 2\alpha - 5 \Rightarrow h'(2) = -1 < 0 \quad (\text{equivalently } h(2 + \epsilon) < h(2)).$$

h is *decreasing* at m , so the minimum lies to the **right**: new interval $[2, 4]$.

Step 2 (iteration 2). Midpoint $m = 3$:

$$h'(3) = +1 > 0 \quad (h(3 + \epsilon) > h(3)) \Rightarrow \text{minimum lies } \mathbf{left}: \text{ new interval } [2, 3].$$

Step 3 (iteration 3). Midpoint $m = 2.5$:

$$h'(2.5) = 0; \text{ numerically } h(2.5 + \epsilon) = h(2.5) + \epsilon^2 > h(2.5) \Rightarrow \text{treated as increasing: new interval } [2, 2.5].$$

Sanity check: the true minimizer $\alpha^* = 2.5$ lies inside every bracket

$$[0, 4] \supset [2, 4] \supset [2, 3] \supset [2, 2.5]. \checkmark$$

Step 4 (width count). Each iteration halves the bracket:

$$\text{width after } k \text{ iterations} = \frac{4}{2^k}.$$

Require $\frac{4}{2^k} \leq 0.01$:

$$2^k \geq 400 \Rightarrow k \geq \log_2 400 \approx 8.64 \Rightarrow k = 9 \quad (\text{width } 4/512 \approx 0.0078).$$

Answer. $[0, 4] \rightarrow [2, 4] \rightarrow [2, 3] \rightarrow [2, 2.5]$; width $= 4/2^k$; $k = 9$ iterations guarantee ≤ 0.01 .

Intuition & exam tips

Fix the decision rule before starting – slope negative $\Rightarrow$ discard the left half, slope positive $\Rightarrow$ discard the right half – and the trace is free marks. The width formula $(b - a)/2^k$ plus one $\log_2$ finishes part (b) in two lines. If the paper swaps in golden-section search, the interval shrinks by factor 0.618 per evaluation instead; if h is not unimodal, the exact bracketing methods fail and the (inexact) Armijo rule is the fallback.

Highest-yield patterns: Problems 6–7 (line-search step on quadratics), 10 (coupled gradients) and 11 (regression by GD) – most exam variants are re-skins of these.